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many more papers in this chapter that are quite useful to readers of *JASA*, including machine design and modeling.

The final chapter contains only three papers, perhaps the most relevant to *JASA* readers being A. Suter's "Outlook for Occupational Noise Regulations in the United States." The author did an admirable job describing the complexities of the U.S. regulation process and, in a sense, predicted the lastest OSHA directive essentially eliminating any citations for A-weighted sound levels below 100 dB, as "time weighted and measured" by OSHA, and this Administration's further use of hearing protectors as the predominant form of worker protection.

In summary then, this two-volume set is valuable because it presents the state-of-the-art. The reader can always review papers for adequacy and call or write the authors for more details. None of the papers were subject to peer review and therefore results and methodology should be checked. The main weakness of the volumes remains a lack of index. I strongly suggest to the editors of future volumes, that they develop at least a subject index to make the book even more valuable as a reference.

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Spatial Hearing: The Psychophysics of Human Sound Localization

Jens Blauert

MIT Press, Cambridge, Massachusetts, 1983.
427 pp. Price $40.00.

Spatial Hearing bridges the gap that separates primarily the psychoacoustician from the telecommunications engineer. Their respective interests, however, meet at the final link in the telecommunications chain. The goal of the engineer is "... to create the auditory illusion that the listener is present at the point of origin of the transmission and directly experiencing the original auditory events ... ." That of the psychologist is to identify and control the acoustic stimuli which influence those location judgments under investigation. Significant progress has been made toward achieving these goals and the author is widely acclaimed as being among the leaders in this quest. Aside from his expertise in electroacoustic systems, Blauert, through his own psychoacoustic research has made fundamental contributions to the literature on localization and lateralization of sound. This then, is the talent and experience behind the writing of *Spatial Hearing.*

The book is partitioned into four chapters. The first three were published originally in German (1974). The introductory chapter sets the rules for the language used in talking about psychoacoustic experiments in general and spatial hearing in particular. It outlines the psychometric methods for obtaining listeners' descriptions of their "auditory events" (Blauert's term), provides an elementary background on sound signals and sound fields, and closes with a brief section on probe microphones, for the author maintains that the most important acoustic input signals to the subjects are the sound signals in the ear canal.

The following two chapters provide basic information on spatial hearing. Chapter 2 deals with the data when only one sound source is present; Chapter 3 treats the data generated by multiple sound sources. Their organization is superb. Blauert first defines localization—the direction *and* distance of an auditory event—and then introduces the term "localization blur"—a concept roughly equivalent to "just noticeable difference." The refinement with which we locate sound and detect small differences in their attributes is conveyed by means of a brief survey of studies published over the past 60 years. Then, for me, the real business of the book begins. Blauert describes the sound field at the two ears resulting from linear distortion caused by the individual elements of the external ear and the head. He presents data on impedance measurements near the ear drum and the resonance frequencies of the pinna. When addressing the effect of the head on the sound field, he provides formal mathematical statements and graphic illustrations of interaural level and phase differences based on the model of the head as a rigid sphere. These are followed by probe microphone measurements designed to obtain the transfer function of the external ear as a function of stimulus frequency with direction of the sound serving as the parameter. After presenting the basic acoustic information, he moves on to the behavioral data. The material is divided into two parts: (1) Evaluating identical ear input signals, and (2) evaluating nonidentical ear input signals. In the former category, directional hearing in the median plane is treated first. Blauert's research on this topic was instrumental in reviving an interest in the pinna's role. He describes his studies and leads the reader toward an appreciation of his concept: directional bands. These are bands of frequency at which sound pressure at the ear drum is greater when the sound originates from certain segments in the median plane *vis-á-vis* other segments. The listener's auditory event when sounds emanate from the median plane is strongly influenced by these directional bands. Studies on auditory distance, a still poorly understood phenomenon, are discussed next, including the case where the sound appears inside the head. One of the more intriguing questions about auditory distance is why we don't hear everything as originating inside our head. Certainly, philosophers long puzzled over our capacity to externalize our visual world. Here, the author lists the conditions necessary for externalizing auditory events and the mystery is somewhat abated.

We evaluate nonidentical ear input signals each time we locate sound binaurally in the horizontal plane. Ironically, the most productive paradigm employed for studying directional hearing in the horizontal plane has been that of lateralization of sound—a procedure which generates an auditory event appearing inside not outside the head. By controlling the signal input at each ear, an enhanced understanding of the influence of interaural time differences and interaural level differences on the lateralization of an auditory event, and presumably directional hearing, is provided. The price for precise control over interaural temporal and level differences is not insignificant; spectral cues for the sound's location provided by the pinna are eliminated since the pinnae are covered by headphones. Chapter 2 is rounded out by presenting data on the effect of head movements on directional hearing in the horizontal plane and, for probably historical purposes, localization theories based on bone conduction, visual, vestibular, and tactile cues are advanced, then quickly demolished as viable contenders for wide acceptance.

The third chapter covers research in which two sound sources radiate either coherent signals or signals with varying degrees of coherency. In the former situation, Blauert covers the studies where signals from two loudspeakers result in an auditory event usually placed somewhere between the two sound sources. He calls this "summing localization" and his selection of the literature reveals the challenging complexity of the phenomenon. Then, the law of first wave front—that which reaches the ear first normally governs perceived direction—is discussed. Fascinating material on the conditions which lead to the emergence of echoes and their perceptual ramifications is presented. When loudspeakers or headphones radiate broadband signals which are more and more incoherent, one consequence is that the auditory image expands. The influence of the degree of signal coherency is discussed in terms of the position and expansion of the auditory event. At this point, binaural masking level differences are treated but their direct relevance to spatial hearing, as Blauert concedes, is questionable. The adequacy by which correlation models of hearing, particularly cross correlation, can account for the perceptual data generated by studies on binaural masking level differences is assessed, and while limited in their applicability, they are nonetheless endorsed. The chapter ends with a consideration of the spatial impressions reported when the sound field is generated in an enclosed space resulting in an array of lateral reflections. Obviously, no "new" auditory events occur as such and Blauert adroitly accounts for the position, extension, and the presence or absence of echoes by means of the summing localization and the law of first wave front. The important practical consideration here is that spatial impression is a primary component of "good acoustics." Blauert describes the electroacoustic methods of achieving the sense of fullness of space at the playback end of the transmission line.

Several years elapsed between the writing of the third and that of the final chapter. New methods of determining the impedance at the ear drum are presented and the transfer characteristics of the external ear are revised in the light of improved measurement techniques. The relatively large amount of research on median plane localization conducted during the 1970's is described leading to an almost inevitable conclusion that median plane localization is governed by the spectral cues provided primarily by the pinna. Lateralization of sound is updated with the phenomenon of multiple images highlighted. Summing localization and the law of first wave front, both important for the understanding of auditory events generated by multiple reflections, are revisited as is the problem of binaural masking level differences. Blauert then summarizes recent advances in binaural signal processing stressing that any credible model must incorporate transfer functions of the external ear, those of the middle ear, frequency selectivity in

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the inner ear, evaluation of interaural time differences, evaluation of interaural level differences, and a comprehensive interpretive treatment of these elements at higher levels of the auditory nervous system. Before finishing, he returns to the topic of spatial impressions. The ability of a telecommunications system to recreate these impressions at the end of the chain is considered the epitome of quality sound reproduction. In this context, Blauert points out the advantages of recording through dummy ears rather than through an array of microphones. When listening through headphones, the listener accepts the reproduction as a valid representation of the original sound field. However, more frequently, playback is through loudspeakers. Techniques which lead to more authentic reproduction of the sounds originally recorded via the dummy head and delivered from loudspeakers have now been developed. Blauert describes them. Thus ends an almost section-by-section account of the text.

In my opinion, the book has two major strengths: (1) localization and lateralization of sound are presented in a beautifully organized and systematic fashion. The mathematics and physics necessary for a general understanding of the physical component of the psychophysical phenomena are incorporated into the discussion. (2) Studies on auditory spatial phenomena are presented which are not in the main stream of psychoacoustic research; yet, the topics form an integral part of our auditory space perception. I am referring to such events as auditory distance and the numerous perceptual consequences of diffuse sound fields. Both receive generous exposure given their limited level of development. Another positive attribute of the book is that the reader is informed of the relevant research written in German; over a third of the references are taken from the German literature. Also, the advances in sound reproduction which are anchored to psychoacoustic research make interesting reading for those with a penchant for applied acoustics. And finally, I should mention the translation. Were it not for the preface to the German edition, I would not have been aware that the first three chapters were originally written in another language, so well did they read.

My main misgiving about the book is that it wasn't rewritten. The author must have had valid reasons for not doing so, but the consequence is a disjointed presentation. There is no escape from the realization that Chap. 4 is tacked on. The topics covered in Chaps. 2 and 3 are again treated in the final chapter. Without question, this last chapter is essential for the vitality of the book. It conveys the sense of continued development of more sophisticated measurement techniques and underscores the richness of the concepts proposed to explain various phenomena of spatial hearing. But, I believe that the book would have been significantly improved had the research over the past 10 years been incorporated into the earlier portions of the text. On a lesser note, I would have preferred more detailed descriptions of several of the figures and an expansion in the description of many of the experimental studies. I also would have taken a more positive approach to some of our exquisite auditory discriminations and not have referred to them as "blurs." These, however, are minor objections and highly idiosyncratic. They should not detract from my strong conviction that *Spatial Hearing* is a fine book!

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